

GRAPH THEORY: NETWORKS & SHORTEST PATH

Lesson Guide & Study Material | The Great Indian Wedding Express!

Scenario: A grand wedding is happening in Delhi! The *Wedding Express* train must rush guests to the right stations on time. But wait... which route is the fastest? Let's use Graph Theory to find out!

1. What are Nodes and Edges?

In graph theory, a **network** is composed of two fundamental building blocks:

Component	Graph Theory Definition	Real-World Example
Node (Vertex)	A distinct point or location in a system.	A railway station, landmark, or market stop.
Edge (Link)	A line or connection between two nodes.	A railway track, road, or metro line.

Together, **Nodes + Edges = NETWORK!**

2. Unweighted vs. Weighted Networks

Unweighted Network	Weighted Network
Edges only show connections (no numbers attached).	Edges have values or numerical weights (e.g., time, distance, or cost).
All connected paths look equal.	Numbers help us determine the quickest or cheapest route.
<i>Example:</i> A simple system map showing which Metro stations connect to each other.	<i>Example:</i> Travel time between stations (e.g., 4 minutes from Rajiv Chowk to Patel Chowk).

3. Exploring Real-World Maps (Delhi Metro Example)

When analyzing a transit network like the Delhi Metro:

- **Stations = Nodes** (places to stop or transfer)
- **Tracks = Edges** (pathways connecting stations)
- **Numbers on tracks = Weights** (travel time in minutes)

4. Algorithm for Finding the Shortest Path

To find the shortest or fastest route in a weighted network, follow these three steps:

1. **Step 1:** List all possible routes from the starting node to the destination node.

2. **Step 2: Add up the travel times** (or edge weights) for each distinct route.

3. **Step 3: Compare the total sums** — the route with the smallest numerical sum is your **shortest path**!

Example 1: Mumbai Dabbawala Route Challenge

The Dabbawalas deliver lunch boxes across Mumbai every day! Here is a delivery network with 5 stops: **A, B, C, D, E**.

Edge weights (travel times):

- $AB = 4 \text{ min}$, $AC = 2 \text{ min}$
- $BD = 5 \text{ min}$, $BE = 8 \text{ min}$
- $CD = 3 \text{ min}$, $DE = 2 \text{ min}$
- $CE = 9 \text{ min}$

Evaluation of routes from A to E:

- Route 1 ($A \rightarrow B \rightarrow E$): $4 + 8 = 12 \text{ min}$
- Route 2 ($A \rightarrow B \rightarrow D \rightarrow E$): $4 + 5 + 2 = 11 \text{ min}$
- Route 3 ($A \rightarrow C \rightarrow E$): $2 + 9 = 11 \text{ min}$
- Route 4 ($A \rightarrow C \rightarrow D \rightarrow E$): $2 + 3 + 2 = 7 \text{ min}$

Result: Route $A \rightarrow C \rightarrow D \rightarrow E$ is the shortest path with a total time of **7 minutes**.

Example 2: School Bus Route Optimiser

Find the fastest route from Home (A) to School (D) via Park (B) or Market (C).

Edge weights: $AB = 4 \text{ min}$, $AC = 7 \text{ min}$, $BD = 5 \text{ min}$, $CD = 3 \text{ min}$, $BC = 2 \text{ min}$

Evaluation of routes from A to D:

- Route 1 (A → B → D): $4 + 5 = 9 \text{ min}$
- Route 2 (A → C → D): $7 + 3 = 10 \text{ min}$
- Route 3 (A → B → C → D): $4 + 2 + 3 = 9 \text{ min}$

Result: Both A → B → D and A → B → C → D tie for the fastest route at 9 minutes.

5. Real-World Applications of Networks

Graph theory and network optimization are applied everywhere in modern transportation and logistics across India:

Jaipur Auto-Rickshaw Routes	Bengaluru Delivery Paths	Indian Railways Network
Delhi Metro Lines	Golden Quadrilateral Highway	Local Bazaar Connections
Mumbai Dabbawala Routes	Traffic vs. Bypass Times	City Road & Market Planning

Key Takeaways

- **Nodes** represent locations; **Edges** represent pathways between locations.
- **Weighted edges** assign numerical values like travel time, cost, or distance.
- The **Shortest Path** minimizes the total weight across all traversed edges.
- Network optimization helps plan efficient real-world transportation, supply chains, and navigation systems.

Practice Questions for Students

Question 1: Terminology Identification

In a city bus routing system, what do the bus stops represent and what do the roads connecting them represent in Graph Theory terms? Explain the difference between an unweighted network and a weighted network in this context.

Question 2: Shortest Path Calculation (Delivery Network)

A delivery driver needs to go from Warehouse (Node S) to Customer (Node T). The available routes and travel times in minutes are:

- $S \rightarrow A = 3 \text{ min}$ | $S \rightarrow B = 6 \text{ min}$
- $A \rightarrow B = 1 \text{ min}$ | $A \rightarrow T = 8 \text{ min}$
- $B \rightarrow T = 3 \text{ min}$

List all possible paths from S to T, calculate the total time for each path, and identify the shortest path.

Question 3: Network Comparison

Suppose you are comparing two metro routes between two distant stations. Route X has 3 intermediate stops with travel times of 5, 4, and 6 minutes. Route Y has only 1 intermediate stop, but due to traffic delays on that line, the two segments take 9 and 7 minutes respectively. Which route is the shortest path in terms of time? Show your work.

Question 4: Real-World Scenario Design

Draw or write down a small network graph representing 4 places in your local neighborhood (e.g., Home, School, Library, Bakery). Assign realistic travel time weights to each edge and demonstrate how you would determine the quickest path from your Home to the Library.